

03 Diversity Indices

EP

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Diversity Indices

Linear parametric models were fit for Richness and Simpson, with each response variable to establish a baseline for future modelling. Species richness, using incidence data, was modelled using a negative binomial GLM. Simpson diversity, using bounded proportion abundance data, was modelled using beta regression with transformation.

```
library(targets)
library(sf)
library(dplyr)
library(betareg)
library(lmtest)
library(statmod)

predictors <- c("n_days", "dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenpace")
```

#If running with downloaded data use:

```
grid_model_all <- st_read("Data/grid_model.gpkg", quiet = TRUE)

df <- grid_model_all %>%
  sf::st_drop_geometry()

richness_models <- readRDS("Data/richness_models.rds")
simpson_models <- readRDS("Data/simpson_models.rds")
```

Uncomment if working from pipeline

```
#setwd("~/CSCT_Pipeline")

#richness_models <- tar_read(richness_models)
#simpson_models <- tar_read(simpson_models)
#grid_model_all <- tar_read(grid_model_all)

#df <- grid_model_all %>%
  #sf::st_drop_geometry()
```

Richness : Negative Binomial GLM

```
rich_null <- richness_models$null
rich_full <- richness_models$full
```

```
summary(rich_null)
```

```
##
## Call:
## glm.nb(formula = sp_richness ~ 1, data = grid_model, init.theta = 0.9742310662,
##       link = log)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.88597    0.04288   113.9  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(0.9742) family taken to be 0.8695553)
##
##      Null deviance: 565.77  on 488  degrees of freedom
## Residual deviance: 565.77  on 488  degrees of freedom
## AIC: 5764
##
## Number of Fisher Scoring iterations: 1
```

```
summary(rich_full)
```

```
##
## Call:
## glm.nb(formula = sp_richness ~ n_days + dtm_elev + avg_imperv +
##       rofsw_risk + rofrs_risk + avg_area_ha + treeht + weighted_precip +
##       weighted_temp + dist_to_river + pct_greenspace, data = grid_model,
##       init.theta = 1.567555521, link = log)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.707e-01  8.546e-01   0.200  0.84181
## n_days         8.613e-04  6.708e-05  12.840  < 2e-16 ***
## dtm_elev       2.989e-04  1.878e-03   0.159  0.87365
## avg_imperv     4.532e-03  2.395e-03   1.892  0.05941 .
## rofsw_riskMedium 1.085e-01  2.166e-01   0.501  0.61697
## rofsw_riskHigh  1.628e-01  1.163e-01   1.400  0.16255
## rofrs_riskLow   -1.486e-01  2.146e-01  -0.692  0.48931
## rofrs_riskMedium -5.616e-02  2.763e-01  -0.203  0.83908
## rofrs_riskHigh  -2.092e-01  2.088e-01  -1.002  0.31725
## avg_area_ha     3.077e-03  3.307e-03   0.931  0.35285
## treeht         -2.713e-02  3.385e-02  -0.801  0.42362
## weighted_precip  3.579e-04  6.620e-04   0.541  0.58913
## weighted_temp   3.738e-01  1.148e-01   3.257  0.00126 **
## dist_to_river   -8.915e-05  1.206e-04  -0.739  0.46020
## pct_greenspace  1.734e+00  3.010e-01   5.759  2.11e-08 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.5676) family taken to be 0.6991593)
##
## Null deviance: 565.23 on 311 degrees of freedom
## Residual deviance: 353.32 on 297 degrees of freedom
## (177 observations deleted due to missingness)
## AIC: 3609.5
##
## Number of Fisher Scoring iterations: 1
```

The full model improves on the null model substantially with both Akaike Information Criterion (AIC) (3609.5 vs 5764) and Residual Deviance (353.32 vs 565.77) reduced. This suggests that the predictors used here have good explanatory power for richness. Three predictors were significant, with positive correlations: `n_days` ($p < 2e-16$), `weighted_temp` ($p = 0.00646$), and `pct_greenpace` ($p = 1.47e-06$). The remaining predictors show no significant linear association with richness in this model. However, the null model is fit on a complete dataset whereas the full dataset is on complete grid squares only (177 rows are dropped due to missingness). Below is a like for like comparison where null is refit to the same conditions as the full model.

```
grid_model_rich <- grid_model_all %>% tidyr::drop_na(all_of(predictors), sp_richness)

rich_null_matched <- MASS::glm.nb(sp_richness ~ 1, data = grid_model_rich)
rich_full_matched <- MASS::glm.nb(
  sp_richness ~ n_days + dtm_elev + avg_imperv + rofsw_risk + rofrs_risk +
    avg_area_ha + treeht + weighted_precip + weighted_temp + dist_to_river +
    pct_greenpace,
  data = grid_model_rich)

lmtest::lrtest(rich_null_matched, rich_full_matched)
```

```
## Likelihood ratio test
##
## Model 1: sp_richness ~ 1
## Model 2: sp_richness ~ n_days + dtm_elev + avg_imperv + rofsw_risk + rofrs_risk +
##          avg_area_ha + treeht + weighted_precip + weighted_temp +
##          dist_to_river + pct_greenpace
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1    2 -1871.6
## 2   16 -1788.8 14  165.75 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This likelihood ratio test confirms that the full model is a significant improvement on the null model ($p < 2.2e-16$) on a matched sample. This indicates the predictor set as a whole explains substantially more variation in species richness than would be expected by chance, providing a solid baseline for future comparison.

Simpson : Beta Regression

```
simp_null <- simpson_models$null
simp_full <- simpson_models$full
```

```
summary(simp_null)
```

```
##
## Call:
## betareg(formula = simpson_adj ~ 1, data = grid_model)
##
## Quantile residuals:
##      Min       1Q   Median       3Q      Max
## -4.9429  0.0376  0.4057  0.5651  1.6158
##
## Coefficients (mean model with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.41800    0.05927   23.92  <2e-16 ***
##
## Phi coefficients (precision model with identity link):
##              Estimate Std. Error z value Pr(>|z|)
## (phi)        2.4800    0.1598   15.52  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Type of estimator: ML (maximum likelihood)
## Log-likelihood: 396.2 on 2 Df
## Number of iterations: 10 (BFGS) + 1 (Fisher scoring)
```

```
summary(simp_full)
```

```
##
## Call:
## betareg(formula = simpson_adj ~ n_days + dtm_elev + avg_imperv + rofsw_risk +
##      rofrs_risk + avg_area_ha + treeht + weighted_precip + weighted_temp +
##      dist_to_river + pct_greenpace, data = grid_model)
##
## Quantile residuals:
##      Min       1Q   Median       3Q      Max
## -5.0548  0.0219  0.3414  0.5709  2.0708
##
## Coefficients (mean model with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    9.877e-01  1.328e+00   0.744  0.4569
## n_days          1.868e-04  1.053e-04   1.774  0.0761 .
## dtm_elev       -5.715e-04  2.915e-03  -0.196  0.8445
## avg_imperv      3.511e-03  3.729e-03   0.942  0.3464
## rofsw_riskMedium 3.086e-01  3.372e-01   0.915  0.3601
## rofsw_riskHigh  1.188e-01  1.813e-01   0.655  0.5125
## rofrs_riskLow   3.949e-01  3.331e-01   1.185  0.2359
## rofrs_riskMedium 4.942e-01  4.302e-01   1.149  0.2507
## rofrs_riskHigh  2.884e-01  3.238e-01   0.891  0.3730
## avg_area_ha     2.917e-03  5.172e-03   0.564  0.5728
## treeht          2.964e-02  5.230e-02   0.567  0.5709
## weighted_precip -6.415e-05  1.030e-03  -0.062  0.9503
## weighted_temp   -2.050e-02  1.784e-01  -0.115  0.9085
## dist_to_river   -1.283e-04  1.874e-04  -0.685  0.4936
```

```
## pct_greenspace      1.019e+00  4.709e-01   2.163   0.0305 *
##
## Phi coefficients (precision model with identity link):
##      Estimate Std. Error z value Pr(>|z|)
## (phi)   2.6526     0.2167  12.24   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Type of estimator: ML (maximum likelihood)
## Log-likelihood: 277.2 on 16 Df
## Pseudo R-squared: 0.08938
## Number of iterations: 25 (BFGS) + 2 (Fisher scoring)
```

The full model has a low psuedo R-squared score (0.08938) and a lower log-likelihood compared to the null model (277.2 vs 396.2) suggesting these predictors have not improved the model despite a small significant result in `pct_greenspace` ($p=0.0305$). Below is a likelihood ratio test to compare the two models.

```
n <- nrow(df)
df$simpson_adj <- (df$simpson * (n - 1) + 0.5) / n
grid_model_simp <- df %>%
  tidyr::drop_na(all_of(predictors), simpson_adj)

simp_null_matched <- betareg::betareg(simpson_adj ~ 1, data = grid_model_simp)
simp_full_matched <- betareg::betareg(
  simpson_adj ~ n_days + dtm_elev + avg_imperv + rofsw_risk + rofrs_risk +
    avg_area_ha + treeht + weighted_precip + weighted_temp + dist_to_river +
    pct_greenspace,
  data = grid_model_simp)

lmtest::lrtest(simp_null_matched, simp_full_matched)
```

```
## Likelihood ratio test
##
## Model 1: simpson_adj ~ 1
## Model 2: simpson_adj ~ n_days + dtm_elev + avg_imperv + rofsw_risk + rofrs_risk +
##      avg_area_ha + treeht + weighted_precip + weighted_temp +
##      dist_to_river + pct_greenspace
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1      2 269.07
## 2     16 277.20 14 16.262    0.2976
```

This shows that, unlike the richness model, the full beta regression model on Simpson diversity does not have a statistically significant improvement on the null model ($p = 0.2976$). This indicates that the predictors have a very low explanatory power for Simpson diversity in a linear model. This is consistent with the pseudo R-squared score (0.08938). It is a weak baseline model but may prove useful when comparing to non-linear models.